INAES - User Manual

Introduction

INAES is an open source software based on R and Shiny, which is a web framework for developing web applications, and can be easily adapted with the user necessity if specific adjustments are needed; however this project aim is to reduce the time spent looking at code, and to help the user to focus more on the data output.
The main sections of INAES software are 7: Freezing Curves, Compare Sample FC, Taxa Analysis, Frozen Fractions, Kneepoint Analysis, Boxplot Analysis and Correlations Analysis.
Each of these analysis has been built based on literature knowledge to adeguate it in the most accurate way to Ice Nucleation theory.

Software Requirements

INAES is developed using R (version 4.3 or higher) and the Shiny framework, an interactive web-based application package in R. Users must ensure compatibility by installing the latest stable version of R (4.3+) and recommended packages listed in the application startup script. The software runs natively within a web browser without additional plugins, ensuring cross-platform usability on Windows, macOS, and Linux systems. INAES can also be deployed locally for offline use.

Data Input Format Specifications

This section provides comprehensive, detailed instructions on how your data must be structured for successful loading and analysis in INAES. Carefully adhering to these formatting rules ensures compatibility and robust functionality across all software modules.

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1. Raw Data File (Droplet Freezing Data)

	Sample.name	Sample_ID	nm	FF	Freezing.temperature	Particle.ratio	nM_corrected	Control	Filtered	Dilution.factor	Location	Size	Sample
1	Sample.name	Sample_ID	nm	FF	Freezing.temperature	Particle.ratio	nM_corrected	Control	Filtered	Dilution.factor	Location	Size	Sample
2	Milliq_L2_1_b5	Milliq_L2_1_b5	329.15	0.22	-21.49	1.00	329.15	Yes	Yes	1	Villum	b_5_m	L2_1
	name on MicroPINGUIN	same but without dilution info like "_b10x"											

File format:

- Acceptable formats: .csv or .txt.
- Column headers must exactly match those specified below (not necessarily in the same order).

Mandatory columns and formatting:

- Sample_ID (string/factor): Unique identifier combining Sample and additional information if necessary (e.g., replicates or experimental runs). Every unique combination of sample properties (e.g., dilution, size) must have a unique Sample_ID. Is basically the name set on the Micro-PINGUIN however with the dilution info removed. (for instance: Sample1_b5_10x becomes Sample1_b5)
- Sample (string/factor): General identifier for the sample. Typically simpler than Sample_ID, often identical to those in the Metadata file. (considering the example above Sample1_b5 should be only Sample1).
- Location (string/factor): Describes geographical origin or experimental location; crucial for grouping and filtering. Entries must be consistent (no typos, consistent casing).
- Size (factor): Particle size fraction labels. Allowed values exactly as follows: b_5_m (for particles <5 µm) or b_02_m (particles <0.2 µm). At the moment deviations or additional sizes are not accepted by the default app logic; in the next update multiple options will be available.
- Dilution.factor (numeric or factor): Dilution level at which the droplets were prepared (e.g., 1, 10, 100, etc.). Consistent numerical entries strongly recommended for robust filtering.
- Freezing.temperature (numeric, °C): Recorded freezing temperature for each droplet (e.g., −12.5). Numeric only; avoid special characters or strings.

- nm (numeric): Calculated ice nucleation particle (INP) concentration at the specific freezing temperature. Values must be numeric, strictly positive (> 0). Missing values must be clearly represented as empty cells or NA.
- FF (numeric between 0–1): Frozen fraction at the corresponding temperature. Defined as the ratio of frozen droplets to total droplets tested at each temperature point. Must be numeric between 0 (no droplets frozen) and 1 (all droplets frozen).
- Control (string/factor): Indicates if the droplet was a control.
  Allowed values exactly as: "Yes" or "No" (capitalized exactly as shown). Controls must be clearly marked to exclude from certain analyses.

Important considerations:
- All numeric values (nm, Freezing.temperature, FF) must be written with dots (.) as decimal separators (e.g., 0.15, -10.5), no commas.
- Ensure that all columns have consistent casing, spelling, and spacing. Any deviation could cause errors or incomplete data loading.

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2. Metadata File

File format:
- Format: .csv (comma-separated only).
- Column headers precisely matching (case-sensitive).

Mandatory columns and formatting:
- Sequencing_ID (string/factor): Unique identifiers matching column headers from the ASV file; critical for correctly mapping sequencing data to sample information. Must be identical character-for-character to column headers in ASV counts (necessary only if you plan to use “Taxa Analysis”).
- Sample (string/factor): The same identifiers used in the Raw Data file to allow linkage of sequencing metadata and freezing data. Must match exactly those listed in the raw data to integrate datasets properly (Sample in Data_raw must be the same as Sample in Metadata).

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3. ASV File (Amplicon Sequence Variant Table)

File format:
• .csv, strictly structured with explicit taxonomy columns.

Mandatory columns and formatting:

Taxonomic columns (must appear first, in this exact order, with exact column names):
- Kingdom (string): Taxonomic kingdom (e.g., Fungi, Bacteria).
- Phylum (string): Taxonomic phylum.
- Class (string): Taxonomic class.
- Order (string): Taxonomic order.
- Family (string): Taxonomic family.
- Genus (string): Taxonomic genus.
- Species (string): Taxonomic species epithet (can be empty or marked as NA if unidentified).
- seq (string): Exact ASV nucleotide sequence used for internal referencing.

These columns must not contain numeric counts—only taxonomy labels or NA for unknown levels.

Sample columns (immediately after taxonomy columns):
- Column headers correspond exactly (character-for-character) to Sequencing_ID entries in the metadata file.

- Each cell must contain numeric values representing raw read counts for the respective ASV in each sequencing run/sample.
- Numeric data only; integers strongly preferred (no decimals).
- Missing or unobserved ASV–sample combinations must be explicitly represented as zero (0).

Important considerations:
- Avoid special characters, spaces, or typos in taxonomic annotations.
- Consistency in naming and capitalization of taxa across the entire file is essential to accurately group ASVs during analysis.
- ASVs without sufficient taxonomic resolution (e.g., missing species) must explicitly contain empty strings or NA in those taxonomic columns.

General Recommendations for File Preparation
- Ensure uniform encoding (UTF-8 recommended) and avoid special hidden characters or formatting marks.
- Remove unnecessary whitespace (leading/trailing spaces in headers or entries).
- Use standard, concise filenames (e.g., freezing_data.csv, metadata.csv, asv_counts.csv) to clearly distinguish data type.

Section 1: Freezing Curves

The Freezing Curves section of the Ice Nucleation Analysis and Exploration Software (INAES) is designed to provide interactive visualization and detailed analysis of droplet freezing data, which are crucial for evaluating ice-nucleating particle (INP) concentrations. This functionality helps investigate the freezing characteristics of samples.

1.1 Overview and Theory

The freezing curve analysis in INAES plots the concentration of ice-nucleating particles (expressed as nM, typically per gram of soil or sample material) against freezing temperatures. The data representation is based on droplet freezing assays, where a series of droplets are cooled gradually, and the freezing events are recorded to derive the INP concentrations at each temperature point.

The plot is structured on a logarithmic scale (log10(nM)), as INP concentrations span multiple orders of magnitude. The logarithmic representation provides clarity in visualizing variations and trends within low- to high-abundance samples.

1.2 User Interface (UI) Description

The Freezing Curves tab features a clear and intuitive layout consisting of a sidebar panel for data filtering, customization, and interaction, alongside a main panel displaying interactive plots.

Sidebar Panel:
- Select particle size (Multi-select dropdown):
- Allows selection of particle size categories (b_5_m, b_02_m) to visualize separately or simultaneously.
- Select dilutions (Multi-select dropdown):
- Filters data points by dilution factors. Dilution filtering enables comparisons across various sample concentrations.
- Select locations (Multi-select dropdown):
- Filters samples according to their geographical or sampling locations. Useful for geographical or environmental comparisons.

Color palette (Dropdown):

- Provides multiple predefined palettes (viridis, magma, plasma, Dark2, Set1, Set2, Set3) for plot colors. Improves visual distinction among different locations or sizes.
- Point border width (Numeric input):
- Adjusts the border thickness of the plot markers. Ideal for enhancing visibility in presentations and publications.

Plot title (Text input):

- Customizes the main title of the freezing curves plot.
- Plot subtitle (Text input):
- Customizes the subtitle for additional descriptive information.

Shape selector (Dynamic selection):

- Dynamically generates options to assign different shapes (Circle, Square, Triangle, Diamond) for each selected location. Shapes aid visual differentiation among categories.

- Show grid lines (Checkbox):
- Enables or disables grid lines for easier data interpretation.
- Update Curves (Button, icon: refresh/sync):
- Applies selected filters and updates the displayed plot accordingly.

Main Panel:

- Interactive Plot (Plotly):
- The freezing curves are displayed interactively, allowing zooming, hovering, and dynamic exploration of data points. Hovering over points reveals detailed information: Particle size, Location, Dilution factor, INP concentration (nm), Freezing temperature (°C), Sample identifier (Sample)

1.3 Functionalities and Workflow

Step-by-step Instructions:

Step 1: Data Upload

- Begin by uploading your experimental dataset in .csv or .txt format using the provided “Upload data file” option on top of the application window.

Step 2: Data Filtering

- Use the sidebar panel to refine your dataset. Select relevant particle sizes, dilutions, and locations of interest.
- Assign visual attributes, such as colors and shapes, to distinguish clearly among sample groups.

Step 3: Visualization

- Click the “Update Curves” button. The software processes and filters your data, applying a 5–95% data trimming to exclude outliers at extreme temperature ranges. This robust statistical method ensures the reliability and interpretability of the displayed curves.

Step 4: Interactive Exploration

- Hover over plot points to view detailed metadata. Interact with the plot to zoom or pan for enhanced data inspection.

Step 5: Plot Customization and Export

- Adjust the aesthetics (palette, shapes, border width) and titles as needed. Use the built-in Plotly tools to export plots in high-resolution SVG format directly from the plot interface. More about plot export in section “Export plots”.

Section 2: Compare Samples FC

The Compare Samples FC (Freezing Curves) module lets you overlay the droplet-freezing profiles of up to three individual samples on a single interactive canvas, so you can inspect similarities or divergences in ice-nucleating behaviour with minimal visual clutter.

2.1 User-Interface

Sidebar controls

- Select up to 3 Samples – an auto-completing field (max 3 entries). The curves you pick here define the comparison set.
- Select particle size – choose one or both size fractions to include.
- Select dilutions – narrow the dataset to specific dilution factors; defaults to all.
- Color palette – identical options to the Freezing Curves tab; governs fill colours in the plot.
- Point border width – numeric spinner (0.1 – 2 px) for marker stroke thickness.
- Plot title / subtitle – free text for figure annotation.
- Shape selector – appears dynamically for each chosen sample; assign Circle, Square, Triangle or Diamond so curves are distinguishable even in monochrome printouts.
- Show grid lines – toggles major/minor grid visibility.
- Update Curves – commits all filter and style changes.

Main panel

- Interactive Plotly canvas – every marker carries hover text: Size, Sample, Dilution, nM, Temperature.

Use native Plotly tools to zoom, pan or export a high-resolution SVG (1600 × 1000 px) via the camera icon.

For the rest, is the same workflow as "Freezing Curves" section.

Section 3: Taxa Analysis

The Taxa Analysis module links molecular-sequencing output (ASV tables) with metadata to explore the taxonomic composition of your samples and to identify taxa that are unique, shared or absent between two groups. All calculations are performed on relative-abundance data derived from your own uploads, so no demo datasets are bundled in the application.

3.1 Required Input Files

1. ASV table (.csv)

Must contain columns named Kingdom, Phylum, Class, Order, Family, Genus, Species, seq followed by one column per sequencing run (the header of those columns should match the values in the metadata column Sequencing_ID).

2. Metadata file (.csv)

Must include columns Sequencing_ID and Sample and may contain additional descriptors (e.g. Location, Latitude).

Upload both files in the header section of the app before opening the Taxa Analysis tab.

3.3 Interface Controls

- Upload ASV table – loads the raw count matrix with taxonomy.

- Select taxonomic level – choose Phylum, Class, Order, Family, Genus or Species for aggregation.
- Analysis mode – pick Per Sample to compare two individual samples or Per Location to compare two geographic pools.
- Dynamic selectors – after the analysis mode is chosen, dropdowns appear so you can pick exactly two samples or two locations.

• In Per Sample mode you choose single samples.
• In Per Location mode you choose locations; the software automatically aggregates all samples that belong to each location.

3.4 Output Panels

- Top taxa lists – two side-by-side DataTables list the most abundant taxa in each selected group.

• Per Sample mode shows the top thirty taxa based on raw counts.
- Per Location mode sums reads across all samples in a location, converts to relative abundance, and shows the top sixty.

• Presence/Absence comparison – an interactive table indicates whether each taxon is: present in both groups; unique to group 1; unique to group 2
A hard abundance threshold of 0.0001 (relative) or 1 read (absolute) is applied; anything below that is treated as absent.
• Bar plot of shared vs unique taxa – a horizontal stacked bar summarises the counts of exclusive and shared taxa between the two groups, with hover text for exact numbers.

All outputs update instantly when you switch samples, locations or taxonomic level.

3.5 Internal Processing Steps

1. Column reconciliation – ASV columns are matched to rows in the metadata via Sequencing_ID and renamed to Sample so downstream plotting uses human-readable names.
2. Filtering of empty columns – sequencing runs that have no mapping in metadata, or that were removed during quality control, are dropped automatically.
3. Taxonomic aggregation – read counts are summed for all ASVs that share the selected rank.

- For Species level the software concatenates Genus + species when both are available, otherwise it falls back to the species epithet alone.

4. Normalisation – in Per Location mode totals are converted to relative abundance to compensate for unequal library sizes.
5. Thresholding – values below 0.0001 relative abundance are set to zero, ensuring that extremely rare OTUs do not inflate the presence/absence matrix.

All steps are executed in reactive expressions, so changes propagate without full app reloads.

Section 4 – Frozen Fraction

The Frozen Fraction tab visualises the cumulative proportion of droplets that have frozen (FF) as a function of temperature for every dilution and particle-size fraction in a single sample.

4.1 Sidebar controls

- Select sample – choose exactly one sample to display.
- Select Size – multi-select of the two standard fractions b_5_m and b_02_m; each selection becomes its own facet.
- Color palette – viridis, magma, or plasma gradients for encoding dilution factor.
- Show control points – include or hide negative-control droplets.
- Shape for sample / control – assign Circle, Square, Triangle or Diamond independently to experimental and control droplets.
- Plot title / subtitle – free-text fields for figure annotation.
- Update FF Plot – triggers data filtering and redraw; also auto-runs on first tab open.

4.2 Main-panel plot (interactive)

- Axes – x-axis: freezing temperature (°C, descending left→right); y-axis: FF (0 – 1).
- Point encoding – fill colour = dilution factor (only if Control == "No"); stroke colour = black, stroke width = 0.3 px; shape is driven by your sidebar choices.
- Facets – one facet per size fraction when multiple sizes are selected; y-scales are independent (scales = "free").
- Hover tooltip – Dilution, Control status, Size, FF, Temperature.
- Export – use the Plotly camera icon for a 1600 × 1000 px SVG (2× scale).

4.3 Internal data pipeline

1. Filtering – rows are subset for the chosen Sample, selected Size values and selected Dilution.factor; control rows may be dropped if the checkbox is off.
2. Colour & shape mapping – a white fill is forced for controls; sample points inherit a continuous palette. Shapes come from scale_shape_manual.
3. Faceting – facet_wrap(~ Size) guarantees curves never overlap across size fractions.
4. Plotly conversion – ggplotly() with WebGL ensures smooth zoom/pan; high-resolution save settings are injected via toImageButtonOptions.

4.4 Recommended workflow

1. Upload the droplet-freezing data and metadata at the top of the app.
2. Open Frozen Fraction and pick the sample you want to inspect.
3. Start with a single size fraction (e.g. b_5_m) plus controls to confirm that blanks remain unfrozen through the test range.
4. Once validated, untick Show control points to declutter the plot.
5. Add the finer fraction (b_02_m) to compare signatures; a left-shifted curve implies highly active sub-micron INPs.
6. Adjust the palette if many dilutions overlap; aim for maximal chromatic separation.
7. Export the plot directly from the mode-bar or copy it via right-click for reports and presentations.

Section 5 – Kneepoint Analysis

(geometric breakpoint detection & statistical validation)

5.1 Theory

In droplet-freezing experiments the log10(nM) curve of a sample often contains one or more distinct “knees”, each theoretically marking the emergence of an independent class of ice-nucleating particles.

The Kneepoint Analysis module implements a piecewise-linear approximation of a smoothed spline and derives rigorous confidence intervals for every breakpoint.

5.2 Mathematical Pipeline

1. Data subset and transformation
- Input filters (Sample, Size, Dilution) restrict the dataset to the curve under investigation.
- Points flagged as controls are dropped.
- The vertical coordinate is converted to y = log10(nM); the horizontal coordinate remains the freezing temperature x (°C).

2. Free spline fit
- A cubic smoothing spline ŷ = s(x; spar) is fit with user-defined smoothness *spar* (0.1–1.5).
- The densely-sampled prediction grid xx (2 000 points) yields {xx, s(xx), se(xx)}.
- The spline captures the overall curve shape while suppressing droplet-level noise.

3. Segmentation into n linear pieces
- A baseline linear model LM_0: $\hat{y} = \beta_0 + \beta_1 x$ is built from the spline points.
- The segmented algorithm searches for ψ1…*n* that minimise RSS when the slope is allowed to change *n* times:

$$\hat{y} = \beta_0 + \beta_1 x + \sum_{k=1}^{n} \gamma_k (x - \psi_k)_+$$

with $(x - \psi_k)_+ = \max(0, x - \psi_k)$

- User control “Number of INP families (breakpoints)” sets n (1–5).
- Convergence is via iteratively-reweighted least squares; it.max = 50 safeguards against non-convergence.

4. Ordering & physical interpretation
- Breakpoints are sorted high → low temperature because higher-temperature knees correspond to stronger (more active) INP classes.
- For each ψk the code finds the closest spline-grid index i and records*
- Tk = ψk [°C]
- nMk = 10ŷi [g^{-1}]*

These pairs form the headline results displayed in the side-panel console.

5.3 Statistical Validation Suite

After clicking Run Statistical Analysis INAES performs four orthogonal checks:

- Parametric CI of ψk – Wald intervals derived from the variance–covariance matrix of the segmented model.
- Sequential F-test (ANOVA vs LM) – compares goodness-of-fit of the segmented model against the single-slope LM_0. A significant p (< 0.05) justifies breakpoint inclusion.
- Bootstrap CI (percentile, R = 500) – non-parametric resampling of the original (x, log nM) pairs; ψk recalculated each iteration. The 2.5 % and 97.5 % percentiles form robust distribution-free bounds.
- 5-fold cross-validation – folds built on the raw curve; breakpoints refit on 80 % of points, predicted on the held-out 20 %. Mean and SD of ψk across folds quantify model stability.

Diagnostic plots (residuals vs fitted and normal Q–Q) are generated for the full segmented model so departures from homoscedasticity or normality are visible at a glance.

5.4 User Controls

- n _breaks_pw – sets the target number of ψ; more breaks increase flexibility but risk over-fitting.
- Spline smoothness (spar) – a lower spar tracks fine-scale wiggles, giving the segmentation more candidate curvature; a higher spar enforces monotone behaviour.
- Display layers – toggle raw points, spline, piecewise fit, breakpoint markers, and ±1.96 SE ribbon for the spline.
- ci_spline_palette / palette_pw – independent colour sets for spline CI and piecewise overlay ensure visual clarity.

Section 6 – Boxplot Comparison

6.1 Theory

The Boxplot Comparison module provides a robust statistical framework for comparing the concentration of ice-nucleating particles (INPs), quantified as nM at −10 °C (nM10) or −15 °C (nM15), across different samples or experimental conditions. This approach helps quantify and visualise variability and significant differences in INP abundance among groups defined by any metadata factor (e.g., geographic location, soil pH, vegetation type, treatment conditions).

6.2 Overview of the Interface

The Boxplot Comparison UI is divided into two primary sections: a sidebar for configuration and a main panel for visualisation and statistical results.

Sidebar Controls

- Select nM variable (Dropdown: nM_10 or nM_15)
  Determines whether the boxplot will focus on INP concentrations at −10 °C or −15 °C.
- Select Size (Dropdown: b_5_m, b_02_m)
  Chooses the particle-size fraction for analysis, either coarse (<5 µm) or fine (<0.2 µm).
- Select variable to compare (Dynamic Dropdown)
  Picks any categorical or numerical metadata column for grouping samples.

Binning method (Radio buttons: Quartiles, Custom number of bins, Custom breaks)
Appears only when a numeric metadata column is selected, offering three approaches:

- Quartiles: divides data evenly into four quantile-based bins.
- Custom number of bins: user-defined number of equal-width bins.
- Custom breaks: manually specified numeric boundaries (comma-separated).
- Generate Boxplot (Button)

Initiates plotting with selected parameters.

- Run Statistical Analysis (Button)
  Performs a series of statistical tests on the generated groups.

6.3 Boxplot Visualisation

The generated boxplot displays the distribution of selected nM values (log-scaled) across defined groups:

- y-axis – log10(nM) with clear power-of-ten annotations (10x format).
- x-axis – selected grouping factor or metadata bins.
- Boxplot elements – medians, quartiles, whiskers, and outliers clearly shown.
- Data overlay – jittered raw data points on top of each box for transparency and individual observation visibility.

Hovering on points displays the Sample ID, facilitating detailed interpretation of individual values.

6.4 Data Pre-processing Steps

1. Data joining

Merges nM values (calculated previously via LOESS regression at fixed temperatures) with user-provided metadata.

2. Grouping creation

For numeric metadata columns, values are discretised using:

- Quartile cuts: dividing continuous data into equal-frequency groups.
- User-defined equal-width intervals.
- Custom-defined numeric boundaries.

For categorical columns, no further discretisation is performed.

3. Filtering

Removes entries with missing values in either the nM measurement or grouping variables, ensuring robust statistical comparison.

6.5 Statistical Validation and Tests

Upon clicking Run Statistical Analysis, the following robust statistical workflow is executed automatically:

Step 1: Normality Testing

- Shapiro–Wilk Test

Evaluates the normality of nM values across groups:

- If $p > 0.05$, data is treated as approximately normal.
- If $p \leq 0.05$, non-parametric tests are preferred.

Step 2: Group Comparisons

- If data are normal:

- ANOVA (Analysis of Variance) is conducted to test overall differences among groups.
- Tukey HSD (Honestly Significant Difference) for pairwise post-hoc tests.
- Eta-squared (η^2) reported for effect-size estimation.
- If data are non-normal:
- Kruskal–Wallis Test evaluates overall differences among medians.
- Dunn's Post-Hoc Test with Bonferroni correction for pairwise comparisons.
- Effect size (Kruskal–Wallis) to quantify the magnitude of observed differences.

Step 3: Diagnostic Plots
- Residual diagnostics:
- Residuals vs. fitted values for variance homogeneity assessment.
- Normal Q–Q plots to visually assess normality.

6.6 Output and Downloads

- Interactive Statistical Summary:
- Main test results displayed clearly in the main panel (p-values, effect sizes).
- Post-hoc comparison results presented in an interactive table (sortable, searchable).
- Bootstrap Confidence Intervals:

Calculated on group means, offering robust, distribution-free confidence intervals (percentile-based from 500 bootstrap samples).
- Downloadable Results:
  A CSV ZIP archive containing detailed results of:
- Main statistical tests (ANOVA/Kruskal–Wallis).
- Post-hoc comparisons (Tukey/Dunn).
- Effect sizes.
- Bootstrap confidence intervals.
- Plot Export:
  Export boxplot figures directly via built-in Plotly controls (high-resolution SVGs).

Section 7 – Correlations Analysis

7.1 Theory

The Correlations Analysis module provides a rigorous statistical framework for evaluating associations between ice-nucleating particle (INP) concentrations at fixed temperatures (nM10 and nM15) and continuous metadata variables (such as latitude, altitude, pH, organic matter content). Understanding these relationships helps reveal underlying environmental drivers and informs hypotheses about the ecological controls on INP distribution and activity.

7.2 User Interface Description

This section is divided into two key panels:

Sidebar Panel
- Analysis Method (Dropdown: Spearman, Pearson, Quadratic Fit, GAM)

Selects the statistical approach used to analyse correlations:
- Spearman/Pearson: linear monotonic relationships.
- Quadratic Fit: nonlinear polynomial relationships.
- Generalized Additive Model (GAM): flexible non-linear relationships.
- Select variable for correlation (Dynamic Dropdown)

Chooses the continuous metadata variable (e.g., latitude, altitude, pH) to be correlated with INP values.

- Run Analysis (Button)
  Executes the selected correlation analysis immediately.
- Download Results (Button)
  Exports high-resolution composite plots containing all correlation analyses performed.

Main Panel

- Two separate plots show correlations between the selected metadata variable and:
- nM10 (INP concentrations at −10 °C)
- nM15 (INP concentrations at −15 °C)

Plots are faceted by particle-size class (b_5_m, b_02_m), each annotated with relevant statistical results (rho, R^2, p-values).

- High-resolution plot download buttons for individual plots:
- Download nM_{10} Plot (PNG)
- Download nM_{15} Plot (PNG)

7.3 Statistical Methods and Interpretation

This module supports four complementary analytical methods:

(1) Spearman Rank Correlation (non-parametric)

- Evaluates monotonic relationships without assuming normality.
- Suitable for datasets with potential outliers or non-linearities.
- Produces correlation coefficient (ρ) and associated p-value.

(2) Pearson Correlation (parametric)

- Assumes linearity, normality, and homoscedasticity.
- Provides correlation coefficient (r) and p-value.
- Sensitive to outliers, recommended for normally-distributed data.

(3) Quadratic Fit (non-linear parametric regression)

- Fits a polynomial of degree two to identify curvilinear relationships:

$$y = \beta_0 + \beta_1 x + \beta_2 x^2 + \epsilon$$

- Model fit evaluated by coefficient of determination (R^2) and significance via an F-test.
- Suitable when a clear curvature is observed (e.g., peaks, thresholds).

(4) Generalized Additive Model (GAM)

- Flexible non-parametric smoothing approach fitting splines (s(x)):

$$y = \beta_0 + s(x) + \epsilon$$

- Ideal for exploring unknown, complex non-linear relationships without strict assumptions about form.
- Reports R^2 and significance of smooth terms.

7.4 Workflow and Computational Pipeline

1. Data Preparation

- Filters metadata entries without nM measurements.
- Merges metadata with nM values calculated via LOESS regression (from earlier sections).

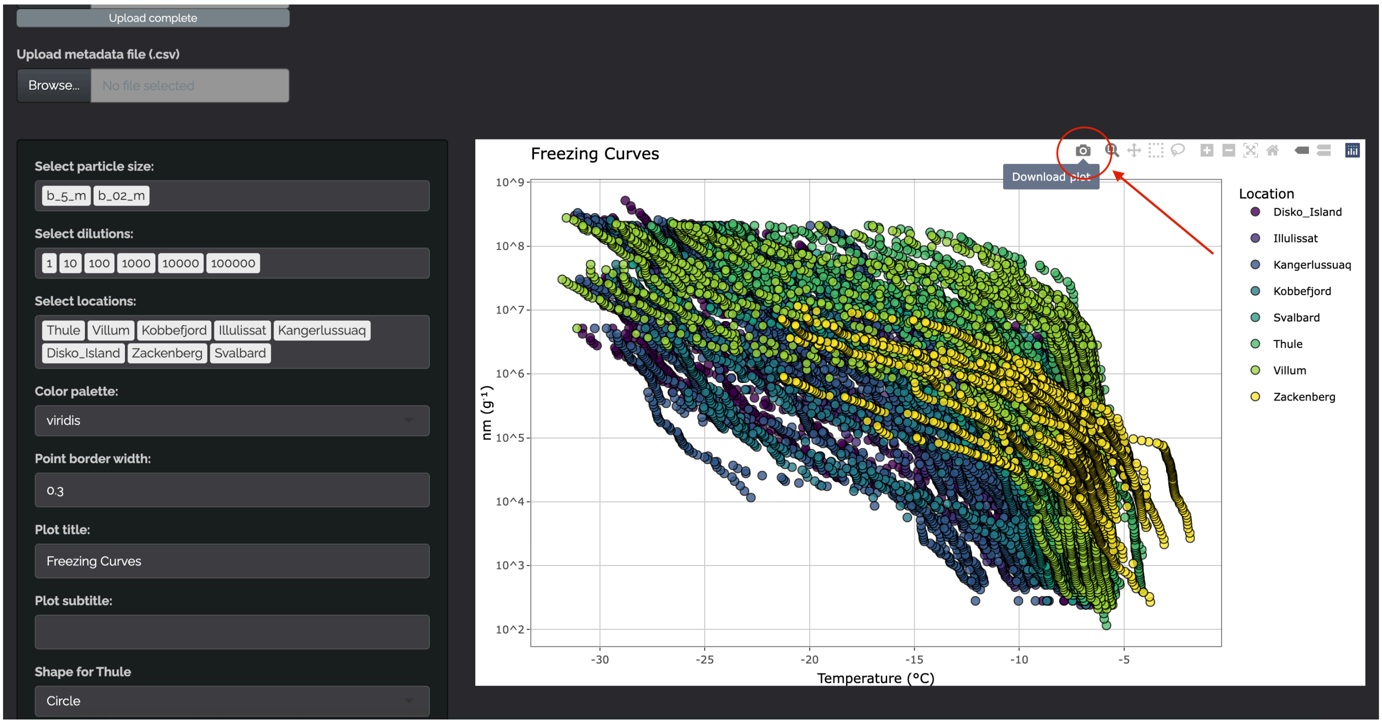
2. Model Fitting (triggered by clicking Run Analysis)
- Automatically executes the selected correlation method separately for each nM metric (10 and 15 °C) and particle-size fraction (b_5_m, b_02_m).
- Reports detailed statistics clearly within plot facets (ρ/r, R^2, p-values).

3. Plot Generation and Labelling
- Plots generated using ggplot2, faceted by size fraction, with logarithmic y-axis (log10(nM)).
- Each facet annotated clearly with statistical results for quick reference and interpretation.
- Smooth curves (regression lines or splines) included with shaded confidence intervals.

4. Reactive Updates and Downloads
- Every run is instantly updated upon method or variable changes.
- Individual high-resolution plots exportable for external reports or presentations.
- Composite plot summarising all correlations downloadable for archival purposes.

Export Plots

To export plots click the picture botton:



If you want to change file type (e.g. from SVG to PNG or others) look into the code for these portions (present for each section):

```
ggplotly(p, tooltip = "text") %>%
    toWebGL() %>%
    config(
      toImageButtonOptions = list(
        format   = "svg",
        filename = "freezing_curves",
        height   = 1200,
        width    = 1800,
        scale    = 2
      )
    )
```

And change ‘format = “svg” ‘ to the wanted file format.